

Notes on Jupyter :

Installation of jupyter needed : "pip install jupyter"

The environment provides a nice editor for programming. It also provides Python for running your code section by section.

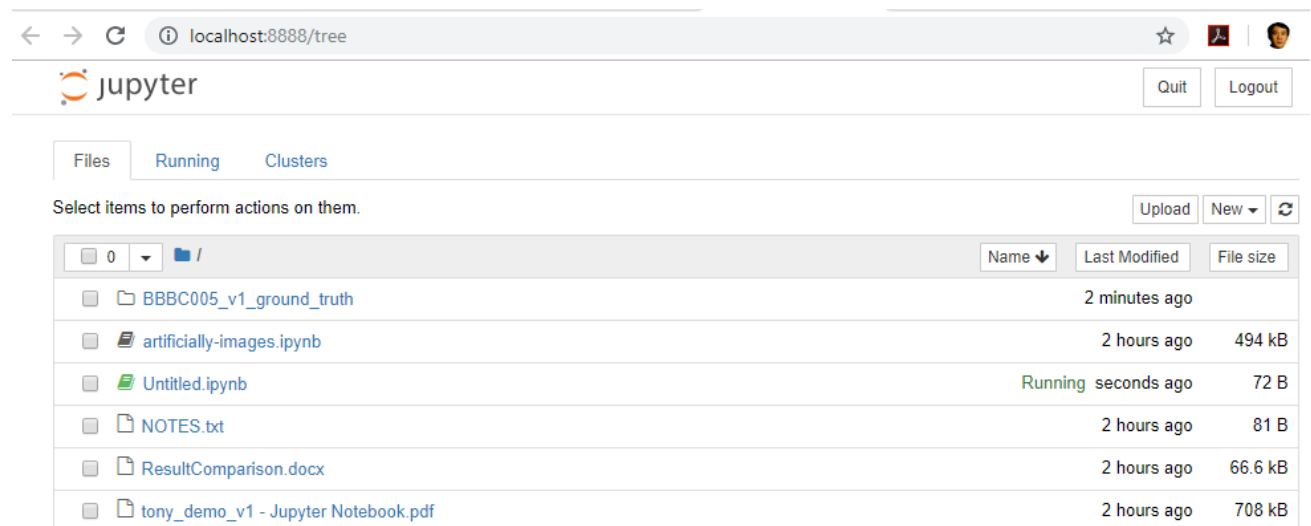
From the **Command Prompt** environment

cd D:

cd <... relevant directory>

D:\gpang> **jupyter notebook**

(Browse to the relevant directory if needed)



Right-Hand-Side → "New" to create a python 3 environment.

- With the new Python3 environment, copy from ***.py** section by section, click "Run"
- After running the code, you can save your work to ***.ipynb**

File → Save as → D:\gpang\.....\code_01.ipynb

```
In [40]: from PIL import Image
import os
import numpy as np
import csv
import pandas as pd

from keras.utils import to_categorical # vectorized the image label
import matplotlib.pyplot as plt
import numpy as np

# File Save as D:\gpgang\...\CaseStudy\code_01.ipynb
# File Save as D:\gpgang\rebirth\Rex_Tony_NDPViewer\Counting_Image_Segmentation\Cell_Counting\CaseStudy

### read training labels ###
os.chdir("D:/gpgang/rebirth/Rex_Tony_NDPViewer/Counting_Image_Segmentation/Cell_Counting/CaseStudy")#plt
df = pd.read_csv('train_label_s30.csv',header = 0)
training_image_name = df['image_name'].values
training_cell_count = df['count'].values
training_blur_level = df['blur'].values
training_stain = df['stain'].values

Using TensorFlow backend.
```

```
In [4]: training_cell_count.shape
training_image_name.shape
```

Out[4]: (30,)

```
In [38]: os.chdir("D:/gpgang/rebirth/Rex_Tony_NDPViewer/Counting_Image_Segmentation/Cell_Counting/CaseStudy/train")
##create a list (training_list) containing pixel value of 2400 training images
training_list = []
for i in training_image_name:
    →im = Image.open(i)
    →#im.show()
    →imarray = np.array(im)
    →training_list.append(imarray)
```

```
In [21]: len(training_list)
training_list[4].shape
```

Out[21]: (520, 696)